



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

FULL SYLLABUS

24-04-2024

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

ANSWER KEY

PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	C	A	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,C	C,D	C,D	B,C	A,C	A,B	A,C	B,C
SECTION-II	Q.	1	2	3	4	5	6		
	A.	4.00	3.00	2.00	2.00	2.00	6.00		

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	C	C	A	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,B,C,D	B,C	C	C	A,B,C,D	A,B,C	A,B,C,D	A,B
SECTION-II	Q.	1	2	3	4	5	6		
	A.	6.00	0.75	4.00	3.00	6.00	0.00		

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	A	D	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	C,D	B,C	A,B	A,B,C	A,C	C,D	A,B,C,D	A,B,C
SECTION-II	Q.	1	2	3	4	5	6		
	A.	1.33	14.00	6.00	1.73	8.00	3.00		

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A)

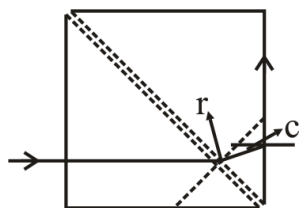
Particle oscillates simple harmonically with

$$\text{time period } T = 2\pi\sqrt{\frac{R}{g}}$$

2. Ans (C)

$$r + c = 60^\circ$$

$$\mu \sin 60 = \sqrt{2} \sin r$$



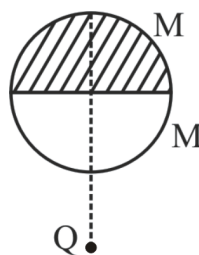
3. Ans (A)

Intensity at Q,

$$\vec{E} = \vec{E}_{\text{upper Hemisphere}} + \vec{E}_{\text{Lower Hemisphere}}$$

$$\frac{G(2M)}{(2R)^2} (-\hat{j}) = \vec{E}_x + -E\hat{j}$$

$$\vec{E}_x = \left[E - \frac{\pi GR\rho}{3} \right] \hat{j}$$



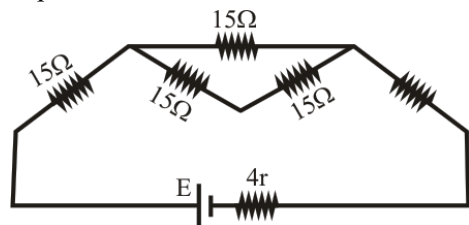
4. **Ans (A)**

Maximum power transfer theorem

Power across load is maximum when external resistance = internal resistance of battery

$$\text{i.e } 40 = 4r \Rightarrow r = 10$$

Equivalent Circuit



PART-1 : PHYSICS

SECTION-I (ii)

5. **Ans (A,C)**

Net force on an element of outer shell

$dF = (dm) [E_s + E] \cos \theta$; $E_s \rightarrow$ intensity due to other elements of shell

$E \rightarrow$ intensity due to inner shell

$$E_s = 2\pi\rho Gt \text{ \& } E = \pi\rho Gt$$

Net force on outer hemispherical shell

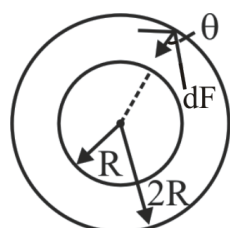
$$F = \int dF \int dm \cos \theta (E_s + E)$$

$$F = \rho (E_s + E) t \pi (2R)^2 \rightarrow (1)$$

By σ in the compressive stress then for equilibrium

$$\sigma 2\pi (2R) t = F \rightarrow (2)$$

$$\text{From (1) \& (2) } \sigma = 2G\rho R t \rho^2$$



6. **Ans (C,D)**

Radius of nucleus $R = R_0 A^{1/3}$, A is mass number

7. **Ans (C,D)**

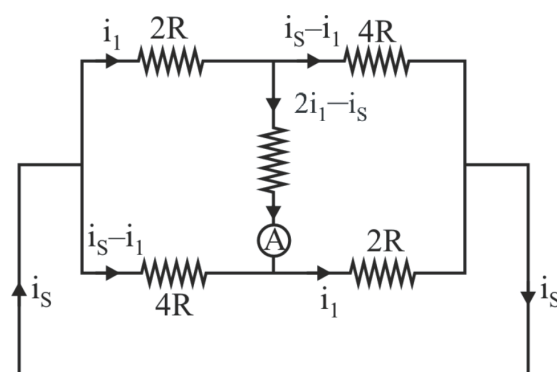
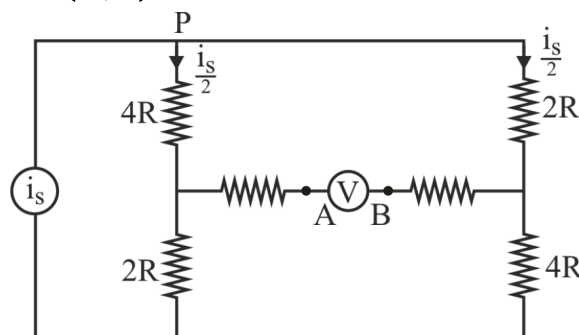
Conceptual

8. **Ans (B,C)**

$$\text{Heat current } H = -KA \frac{dT}{dx}$$

$$\text{Slope of } TV/S \text{ x graph } \frac{dT}{dx} = \frac{-H}{KA}$$

9. **Ans (A,C)**



$$V_P - V_A = 2i_s R$$

$$V_B - V_A = i_s R$$

$$V_P - V_B = i_s R$$

From KCL

$$2i_1 - i_s = \frac{-i_s}{9}$$

10. **Ans (A,B)**

Conceptual

11. **Ans (A,C)**

Thrust force $\rho A v^2 = mg$, for equilibrium

$$v = \frac{1}{r} \sqrt{\frac{mg}{\pi\rho}}$$

$$\begin{aligned} \text{Power delivered by engine} &= \frac{dK}{dt} = \frac{1}{2} \frac{dm}{dt} v^2 \\ &= \frac{mg}{2r} \sqrt{\frac{mg}{\pi\rho}} \end{aligned}$$

12. **Ans (B,C)**

From continuity equation $Q = \rho (2\pi r \ell) v$; $v \propto \frac{1}{r}$

Also from bernoulli's theorem $P + \frac{1}{2} \rho v^2 = \text{const}$

$$\text{Hence } \Delta P \propto \frac{1}{r^2}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (4.00)**

Heat generated is same in all frames

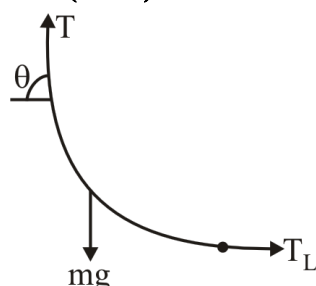
$$W_{\text{external forces}} + W_{\text{pseudo}} = \Delta U_{\text{elastic energy}} + H \left[\text{in C.O.M frame} \right]$$

$$W_{\text{ext}} = \frac{3F}{2} \Delta \ell \text{ where } \Delta \ell = \frac{3F \ell}{2AY}$$

$$\Delta U = \frac{7}{6} \frac{F^2 \ell}{AY};$$

$$\text{heat generated } H = \frac{13F^2 \ell}{12AY} \quad a = 1; b = 3$$

2. **Ans (3.00)**



Equilibrium conditions

$$T \cos \theta = T_L = 90$$

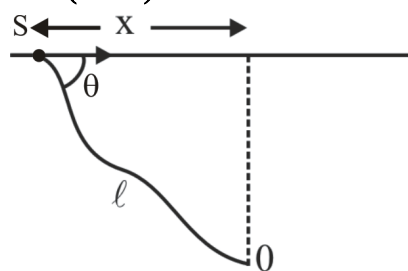
$$T \sin \theta = mg$$

$$T \tan \theta = \frac{mg}{90}$$

$$\sec^2 \theta d\theta = \frac{(dm)}{90} g \text{ where } dm = \mu (R d\theta)$$

$$R = 3m$$

3. **Ans (2.00)**



Apparent frequency

$$f^1 = f \frac{V}{V - V_S \cos \theta}$$

$$\text{But } \cos \theta = \frac{x}{\ell} = \frac{V_S}{V}$$

$$f^1 = \frac{fV^2}{V^2 - V_S^2}; \text{ wavelength recorded}$$

$$\lambda^1 = \frac{V}{f^1} = \frac{2}{11} m$$

4. **Ans (2.00)**

When the drop attains terminal speed

$$\frac{dm}{dt} v = mg \text{ but } \frac{dm}{dt} = Kmv$$

$$\text{Hence } v = \sqrt{\frac{g}{K}} = 2 \text{ m/s}$$

Ans (2.00)

$$\left\langle \frac{dv}{dx} \right\rangle = \left\langle \frac{dK}{dx} \right\rangle = \frac{1}{4} \mu A^2 \omega^2$$

$$\left\langle \frac{dv}{dx} \right\rangle = \frac{1}{2} \mu A^2 \omega^2 \text{ hence energy over due}$$

wavelength

$$E = \left\langle \frac{dE}{dx} \right\rangle \lambda$$

6. **Ans (6.00)**

$$C = \frac{\Delta Q}{\Delta T} = C_V + \frac{W}{\Delta T} = \frac{3R}{2} + \frac{W}{\Delta T}$$

$$\frac{W}{\Delta T} = \frac{2(n+3)}{(5n-3)}$$

$$\frac{3R}{2} + \frac{2(n+3)}{(5n-3)} = \frac{13R}{6} \Rightarrow n = 6$$

PART-2 : CHEMISTRY

SECTION-I (i)

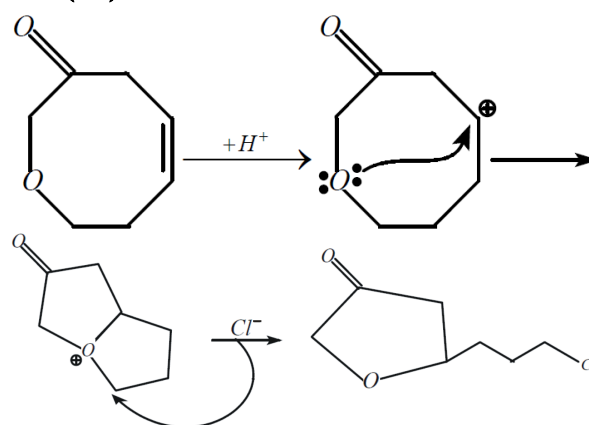
1. **Ans (C)**

A) Trans major

B) Trans major

D) Trans major

2. **Ans (C)**

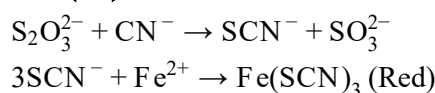


3. **Ans (A)**

$$\Delta G = -2.303nRT \log \frac{P_1}{P_2}$$

$$= -2.303 \times 2 \times 8.314 \times 273 \log \left(\frac{2}{1} \right) = -3147J$$

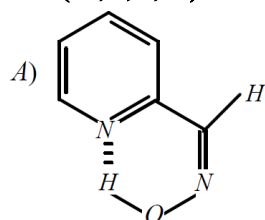
4. **Ans (B)**



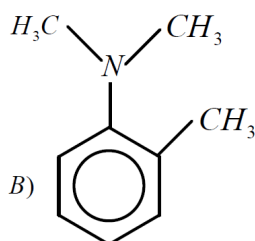
PART-2 : CHEMISTRY

SECTION-I (ii)

5. **Ans (A,B,C,D)**



Since l.p. involved in H - bonding, it less basic than its anti-isomer



Due to S.I.R, l.p. at N not involved in delocalization and hence it is more basic than Dimethylaniline

C) From molecular orbital calculations, loss of proton is 5 K.cal/mole easier for the syn conformation than anti formation. In acyclic molecules preferred conformation is anti but in cyclic molecule preferred conformation is syn

D) + I - effect $(\text{CH}_3)_3\text{C} - < (\text{CH}_3)_3\text{Si} -$ due to less E.N. of Si than carbon

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A)**

Conceptual

2. **Ans (A)**

$$I = \int_0^{\frac{1}{2}} \sin 4\pi x g(x) dx + \int_{\frac{1}{2}}^1 \sin 4\pi x g(x) dx$$

(I₁) (I₂)

For I₂, put $x = \frac{1}{2} + t$

3. **Ans (D)**

Use triangle inequality

4. **Ans (B)**

Take cross product with $\bar{p}$ to set $\bar{r}$ and take dot product with $\bar{p}$ to set λ , take dot product with $\bar{q}$ to get $[\bar{p}\bar{q}\bar{r}]$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (C,D)**

$$2sp.s^1p = (sp + s^1p)^2 - ps^2 - p^1s^2$$

$$= 2a^2 + 2b^2 - 2x_1^2 - 2y_1^2$$

$$sp.s^1p = d^2 = a^2b^2$$

7. **Ans (A,B)**

Using algebra of limit

8. **Ans (A,B,C)**

$$P(E_1) = \frac{6c_3}{6^3} \quad P(E_1 \cap E_2) = \frac{1}{36}$$

$$P(E_2) = \frac{6}{6^3}$$

9. **Ans (A,C)**

$$f(x) = \frac{(x^4 + x^5 + x^6)^7}{7}$$

10. **Ans (C,D)**

Normal at 't' is

$$y + xt = 2at + at^3$$

(3, 0) Lies on at $\Rightarrow t = 0, 1, -1$

P = (1, 2), Q = (0, 0), R = (1, -2)

11. **Ans (A,B,C,D)**

$$f''(x) = \lambda x \Rightarrow f'(x) = \frac{\lambda x^2}{2} + c$$

$$\Rightarrow f'(1) = 0 \Rightarrow c = \frac{-\lambda}{2}$$

$$f'(x) = \frac{\lambda}{2}(x^2 - 1) \Rightarrow f(x) = \frac{\lambda}{2} \left(\frac{x^3}{3} - x \right) + c$$

$$f(x) = \frac{-x^3}{2} + \frac{3x}{2} + 1$$

12. **Ans (A,B,C)**

$\sin x = 1$ Gives one solution and $\sin x = \alpha$ gives

other solution such that $\alpha > 1$ or $\alpha \leq 0$

$\alpha \leq 0 \Rightarrow 1 + b = a$ And $b > 1$ or $b \leq 0$

PART-3 : MATHEMATICS**SECTION-II**1. **Ans (1.33)**

Tangent at A will be perpendicular to BC so,

$$\text{slope of tangent at A} = -\frac{1}{2}$$

$$\alpha = \pm\sqrt{2}, \beta = \pm 2\sqrt{2}$$

2. **Ans (14.00)**

$$P = 1 - \frac{{}^6C_1 \cdot 5^4}{6^5}$$

$$= \frac{6^4 - 5^4}{6^4}$$

3. **Ans (6.00)**

$$A^{-1} B B B^{-1} + A B B^{-1} = 0$$

$$\Rightarrow B = -A^2$$

$$\text{and } |A| = 5$$

$$A^6 - 2A^4B + A^2B^2 = 4A^6$$

4. **Ans (1.73)**

$$z = -1, 1, w, w^2$$

5. **Ans (8.00)**

Equation of L are angular

Bisection of L_1 and L_2 6. **Ans (3.00)**Use $AM = BM = PM$

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

ANSWER KEY

PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	B,D	A,C	A,C	A,C	A,B,D	A,C	B,D	A,B,C
SECTION-I (ii)	Q.	9	10	11	12				
	A.	A	A	C	A				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	8.00	5.00	1.00	5.00	7.00	4.40		

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C	B	A,B	B,C,D	A,C,D	A,C,D	A,B,C	A,B,C,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	A	A	D	A				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	69.40	12.00	2.67	18.60 to 18.80	7.00	120.00		

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	B,C	A,B,D	D	A,C,D	A,C	B,C	A,B,C	A,B,C
SECTION-I (ii)	Q.	9	10	11	12				
	A.	C	C	C	A				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	2021.00	0.50	8.00	4.00	3.00	2.00		

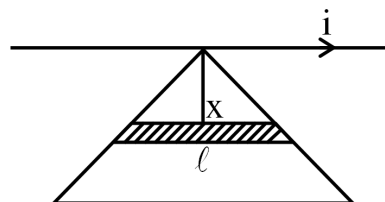
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PART-1 : PHYSICS
SECTION-I (i)

 1. **Ans (B,D)**

Conceptual

 2. **Ans (A,C)**

From Lenz's law current in infinite wire (loop) is anticlockwise and hence to the right in the wire, leading to a repulsive force on loop.



Flux across loop is

$$\phi = \int_0^h \frac{\mu_0 i}{2\pi x} \ell dx = \frac{\mu_0 i b}{2\pi} = \frac{\mu_0 i h}{\pi}$$

$$\text{Mutual inductance, } M = \frac{\mu_0 h}{2\pi}$$

$$\text{Induced emf } e = M \frac{di}{dt} = \frac{\mu_0}{\pi}$$

3. **Ans (A,C)**

If earth is sphere then, $E = \frac{GM}{r^2} = \frac{4}{3} \frac{\pi \rho G R^3}{r^2}$

If earth is cylindrical then,

$$E = \frac{2G\lambda}{r} = \frac{2G\rho\pi R^2}{r}$$

If E increases then 'v' increases.

4. **Ans (A,C)**

$$i = \frac{dQ}{dt} = \frac{d}{dt} \left(\frac{\epsilon_0 A}{d} V \right) = \frac{\epsilon_0 \omega A \alpha V}{(d_0 + \alpha \cos \omega t)^2} \sin \omega t$$

5. **Ans (A,B,D)**

Work is done by Electric field only

$$W_E = \frac{1}{2} m [(2V)^2 - (V)^2] = \frac{1}{2} m 3V^2$$

$$qE(2a) = \frac{3mV^2}{2}; E = \frac{3mV^2}{4qa}$$

$$\text{Power delivered by a force} = \vec{F} \cdot \vec{V}$$

6. **Ans (A,C)**

Potential due to the dipole and the induced charges on the inner surface of cavity is zero everywhere outside cavity.

7. **Ans (B,D)**

$$\text{Force exerted on the wall, } F = 2\rho A v^2$$

8. **Ans (A,B,C)**

Current flows only in the parallel branch of the secondary circuit.

PART-1 : PHYSICS

SECTION-I (ii)

9. **Ans (A)**

Conceptual.

10. **Ans (A)**

Conceptual.

11. **Ans (C)**

Conceptual.

12. **Ans (A)**

Conceptual.

PART-1 : PHYSICS

SECTION-II

1. **Ans (8.00)**

$$\text{Apparent frequency, } f^1 = \frac{fV^2}{V^2 - V_S^2} \text{ when}$$

source is closest, $\frac{V_S}{V} = \sqrt{\frac{2}{3}}$, distance of source when original frequency is recorded

$$= \sqrt{d^2 + \left(V_S \frac{d}{V} \right)^2} = d\sqrt{\frac{5}{3}}, \text{ where } d = 300\text{m}$$

$$x + y = 15.33.$$

2. **Ans (5.00)**

At terminal speed; upthrust = Viscous force,

$$\rho \frac{4}{3} \pi r^3 g = 6\pi \eta r V; V = 4 \text{ cm/s,}$$

$$\text{time} = \frac{20}{4} = 5 \text{ sec}$$

3. **Ans (1.00)**

$$E_1 = mS\Delta\theta = (\rho A \ell) S \left(\frac{\Delta \ell}{\alpha \ell} \right) = 180\text{J}$$

$$E_2 = \frac{1}{2} Y (\text{Strain})^2 A \ell = \frac{9}{5} \text{J}; \frac{E_1}{E_2} = 100.$$

4. **Ans (5.00)**

$$\Delta Q = \Delta W; \Delta W = \Delta W_{23} + \Delta W_{31} + \Delta W_{12}$$

$$\Delta W = 1250 + (-500) + 0 = 750$$

$$\text{Also } P_2^{1 \rightarrow 2} T_2^1 = P_3^{1 \rightarrow 2} T_3^1; T_3 = 450\text{K.}$$

5. **Ans (7.00)**

Heat transfer rate across an elementary shell of radius 'r' and thickness 'dr' is given by

$$\frac{\Delta Q}{\Delta t} = P \frac{r^3}{R^3}; \text{ i. e., } -K4\pi r^2 \frac{dT}{dr} = P \frac{r^3}{R^3}$$

$$- \int_{T_C}^{T_S} 4\pi R^3 K dT = P \int_0^R r dr \text{ but } P = \sigma 4\pi R^2 T_S^4 \text{ for equilibrium.}$$

$$-(T_S - T_C) = \frac{\sigma T_S^4 R}{2K}; \text{ Hence } m = 1, n = 4, \& \ell = 2.$$

6. **Ans (4.40)**

$$U_i + K_\alpha = U_f + K(m + M) \dots (1)$$

$$\sqrt{2m_\alpha K_\alpha} = \sqrt{2(m + M) K_{m+M}} \dots (2)$$

$$K_\alpha = \frac{m + M}{M} |Q| = 4.4 \text{ MeV.}$$

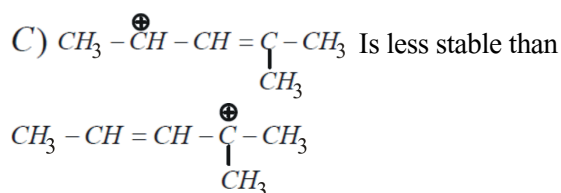
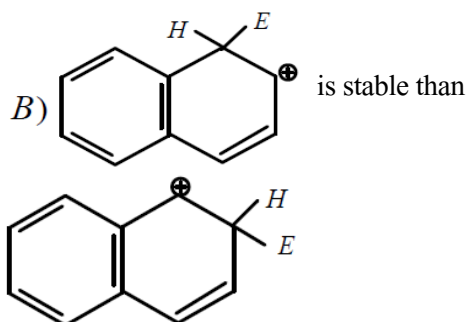
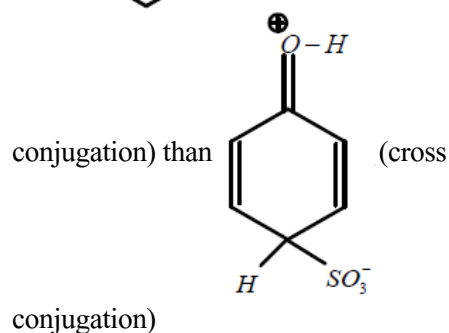
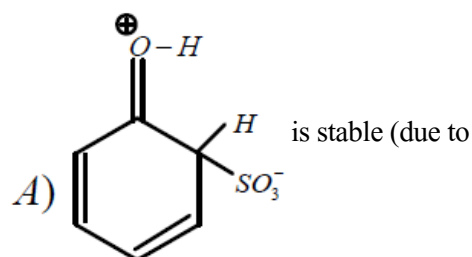
PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (A,B,C)**

If the atom that is forming the new bond to carbon is the same over a range of nucleophiles, then nucleophilicity does parallel to basicity
 Acidic strength ; $\text{H}_2\text{O} < \text{PhOH} < \text{AcOH} < \text{TsOH}$
 Basic strength; $\text{OH}^- > \text{PhO}^- > \text{ACO}^- > \text{TsO}^-$
 down the group nucleophilicity increases
 $\text{PhO}^- < \text{PhS}^-$

3. **Ans (A,B)**



D) At carbonyl carbon electrophilicity more than at β - carbon. $\text{C}=\text{O}$ B.E is more than the $\text{C}=\text{C}$ bond energy. Thus carbonyl addition product is K.C.P, conjugate addition product is T.C.P

7. **Ans (A,B,C)**

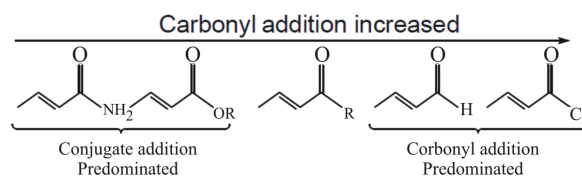
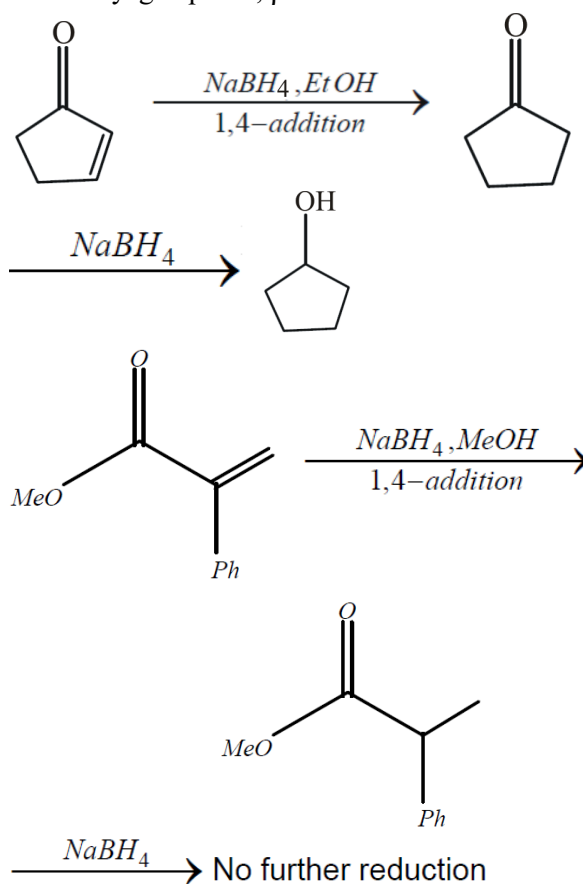
'C' and 'D' reactions take place during the smelting process

PART-2 : CHEMISTRY

SECTION-I (ii)

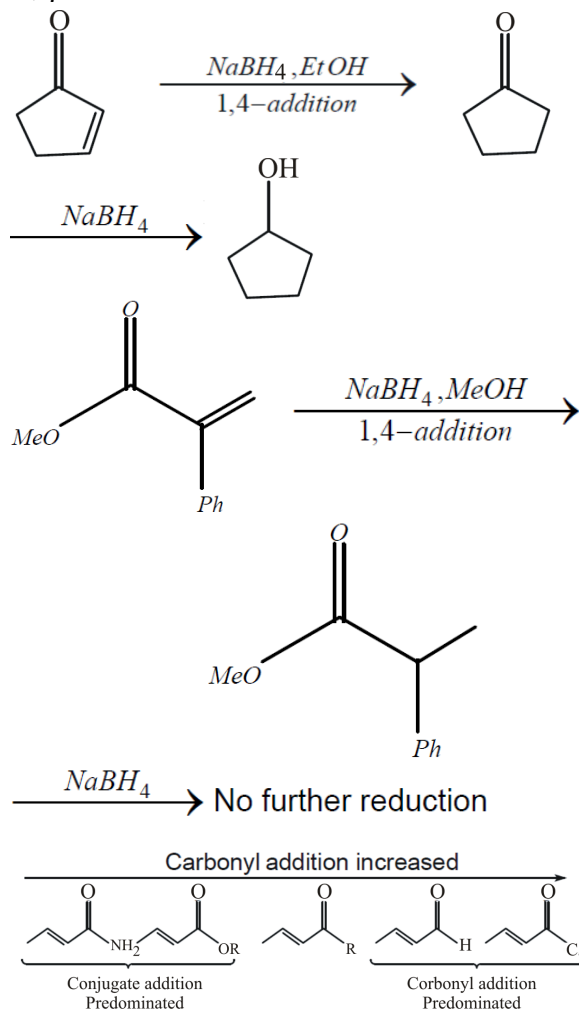
9. **Ans (A)**

Correct match is I – P, II – Q, III – R, IV – U.
 Reduction with NaBH_4 in the presence of CeCl_3 is called Luche reduction. It causes selective reduction of carbonyl group in α, β - unsaturated ketones



10. Ans (A)

Correct match is I – P, II – Q, III – R, IV – U.
Reduction with NaBH_4 in the presence of CeCl_3 is called Luche reduction. It causes selective reduction of carbonyl group in α, β – unsaturated ketones



11. Ans (D)

Correct match is

I – PQS

II – PRS

III – PRSU

IV – RSTU

In adiabatic free expansion of real gases some translational kinetic energy changes to potential energy in such a way that total internal energy remains constant. Due to Change in K.E, temperature changes. If $T < T_i$, $\Delta T - V_e$, if $T > T_i$, $\Delta T + V_e$

12. Ans (A)

Correct match is

I – PQS

II – PRS

III – PRSU

IV – RSTU

In adiabatic free expansion of real gases some translational kinetic energy changes to potential energy in such a way that total internal energy remains constant. Due to Change in K.E, temperature changes. If $T < T_i$, $\Delta T - V_e$, if $T > T_i$, $\Delta T + V_e$

PART-2 : CHEMISTRY

SECTION-II

1. Ans (69.40)

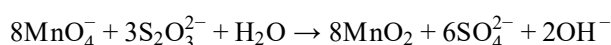
No. of moles of organic compound = $1.34 / 134 = 0.01$

No. of moles of AgI precipitated = $7.05 / 235 = 0.03$

$\therefore$ no. of methoxy groups per molecule = $0.03 / 0.01 = 3$

W% of CH_3O – groups = $\frac{31 \times 3}{134} \times 100 = 69.4\%$

3. Ans (2.67)



$$-\frac{1}{8} \frac{d[\text{MnO}_4^-]}{dt} = \frac{-1}{3} \frac{d[\text{S}_2\text{O}_3^{2-}]}{dt}$$

$$-\frac{d[\text{MnO}_4^-]}{dt} = \frac{8}{3} \left(\frac{-d[\text{S}_2\text{O}_3^{2-}]}{dt} \right)$$

4. Ans (18.60 to 18.80)

$$d_{(\text{hkl})} = \frac{a}{\sqrt{h^2 + K^2 + l^2}} \Rightarrow 100 = \frac{a}{\sqrt{2^2 + 2^2 + 2^2}}$$

$$\Rightarrow a = 100 \times 2\sqrt{3} = 200 \times 1.732$$

$$= 346.4 \text{ pm}$$

5. **Ans (7.00)**

$$1.4 \times 10^{-4} = \frac{[\text{Ag}(\text{NH}_3)^+][\text{NH}_3]}{[\text{Ag}(\text{NH}_3)_2^+]}$$

$$[\text{Ag}(\text{NH}_3)^+] = \frac{1.4 \times 10^{-4} \times 10^{-3}}{2 \times 10^{-1}}$$

$$= 0.7 \times 10^{-6} = 7 \times 10^{-7} \text{ M}$$

6. **Ans (120.00)**

$$c = b^2 = 16 \times 10^{-4} \Rightarrow b = 0.4 \text{ } \ell\text{t/mole}$$

$$V_C = 3b = 3 \times 0.04 = 0.12 \text{ } \ell\text{t / mole}$$

$$= 120 \text{ cm}^3/\text{mole}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (B,C)**

For minimum area A', B' and C' must be mid

points of QR, PR & PQ respectively,

$$\therefore A'(0, 3, 4), Q(0,6,0), R(0,0,8), C'(2,3,0)$$

$$P(4,0,0), B(0,3,0)$$

$$\text{Volume} = \frac{1}{6}(4 \times 6 \times 8) = 32$$

Shortest distance is OB i.e. 3

2. **Ans (A,B,D)**

Use 'V_n' method $K = 1/840$

3. **Ans (D)**

When each element of any row (column) of a determinant is the sum of two terms, then it can be expressed as the sum of two determinants of the same order

4. **Ans (A,C,D)**

$g(x)$ is discontinuous if $c \neq 4$.

N_1 : Given both $f(x)$ and $g(x)$ are discontinuous

Hence, $a \rightarrow 6$ ways $\therefore N_1 = 6 \times 5 = 30$

$c \rightarrow 5$ ways

N_2 since $g(x)$ is discontinuous $\therefore c \neq 4$.

Now, $h(x)$ is continuous at $x = 1$.

$$h(1^-) = h(1^+) \Rightarrow 2(1 - c) = (1 - a)(1 - c)$$

$$\Rightarrow (1 - c)(1 + a) = 0$$

$$\therefore 1 - c = 0 \text{ (} a \neq -1 \text{) (Given)}$$

$$\Rightarrow c = 1$$

Also, given $h(x)$ is discontinuous at $x = 2$.

$$(4 - 2a)(4 - 2c) \neq (4 - 2a)(-4)$$

$$(4 - 2a)(8 - 2c) \neq 0$$

$$(2 - a)(4 - c) \neq 0$$

Since $c = 1$

$$\therefore a \neq 2 \Rightarrow a \rightarrow 5 \text{ ways}$$

$c \rightarrow 1$ way

$$\therefore N_2 = 5$$

Now, N_3 given $h(x)$ is discontinuous at $x = 1$.

$$\Rightarrow (1 - c)(a + 1) \neq 0 \therefore c \neq 1$$

and given $h(x)$ is continuous at $x = 2$

$$\Rightarrow (2 - a)(4 - c) = 0$$

if $c = 4$ then $a \rightarrow 6$ ways

if $a = 2$ then $c \rightarrow 4$ ways

Total 10 ways.

$$\therefore N_1 = 30$$

$$N_2 = 5$$

$$N_3 = 10$$

5. Ans (A,C)

$$\because g'(x) = 3x^2 + 2x + 0 > 0, \forall x \geq 0$$

$\Rightarrow g(x)$ is an $\uparrow$ ing function.

If circumcenter lies outside, then triangle is obtuse angle

triangle and angle containing the given sides is obtuse angle.

Therefore,

$$\left(f(x)\hat{i} + g(x)\hat{j} \right) \cdot \left(g(x)\hat{i} + f(x)\hat{j} \right) < 0$$

$$\Rightarrow f(x) \cdot g(x) < 0 \quad \dots\dots (i)$$

$$\Rightarrow g(x) \uparrow \text{ for } x \geq 0$$

$$\Rightarrow g(x) > g(0) \quad \forall x > 0, \text{ Also, } g(0) = 0$$

$$\Rightarrow g(x) > 0 \quad \forall x > 0, (i) \Rightarrow f(x) < 0$$

$$\therefore f(x) < 0 \text{ and } g(x) > 0 \quad \forall x > 0$$

If $x \rightarrow \infty$ then $g(x) \rightarrow \infty$ and $f(x)$ is some negative number,

then

$$\lim_{t \rightarrow 0} \lim_{x \rightarrow \infty} \left(\underbrace{\cot \left(\underbrace{\frac{\pi}{4} (1-t^2)}_{1^-} \right)}_{1^+} \right)^{\underbrace{\frac{f(x) \cdot g(x)}{t \rightarrow \infty}}_{\infty}} = 0$$

6. Ans (B,C)

$$P(A) = 6 \text{ or } \overline{6} \overline{7} \overline{6} \text{ or } \overline{6} \overline{7} \overline{6} \overline{7} \text{ or } \dots\dots \infty$$

$$= \frac{5}{36} + \frac{31}{36} \times \frac{30}{36} \times \frac{5}{36} + \left(\frac{31}{36} \times \frac{30}{36} \right)^2 \times \frac{5}{36} + \dots\dots \infty = \frac{30}{61}$$

$$P(B) = \frac{31}{36} \times \frac{6}{36} + \frac{31}{36} \times \left(\frac{30}{36} \times \frac{31}{36} \right) \times \frac{6}{36} + \frac{31}{36} \times \left(\frac{30}{36} \times \frac{31}{36} \right)^2 \times \frac{6}{36} + \dots\dots \infty = \frac{31}{61}$$

7. Ans (A,B,C)

$$\Rightarrow f'(x) = e^{\tan x} \cdot (1 - \sin 2x) \geq 0 \quad \forall x \in \left(\frac{-\pi}{2}, \frac{\pi}{2} \right)$$

$$\Rightarrow f(x) \uparrow \text{ on } \left(\frac{-\pi}{2}, \frac{\pi}{2} \right)$$

$$\Rightarrow f''(x) = e^{\tan x} \cdot (-2 \cos 2x) + e^{\tan x} \cdot \sec^2 x \cdot (1 - \sin 2x)$$

$$f''(x) = e^{\tan x} \left(\frac{2(\tan^2 x - 1)}{1 + \tan^2 x} + (\sec^2 x - 2 \tan x) \right)$$

$$= f(x) \cdot (\tan x - 1)(\tan^3 x - \tan^2 x + 3 \tan x + 1)$$

$$\text{Now, let } g(x) = \tan^3 x - \tan^2 x + 3 \tan x + 1$$

$$\Rightarrow g'(x) = (3 \tan^2 x - 2 \tan x + 3) \sec^2 x > 0 \quad \forall$$

$$x \in \left(\frac{-\pi}{2}, \frac{\pi}{2} \right)$$

So, $g(x)$ is increasing function.

$$\text{Also, } g\left(\frac{-\pi}{4}\right) g(0) < 0$$

$$\text{So, } g(x) = 0 \text{ has exactly one root in } \left(\frac{-\pi}{2}, \frac{\pi}{2} \right),$$

where one root is $\frac{\pi}{4}$ and other root lies in

$$\left(\frac{-\pi}{4}, 0 \right).$$

$$\text{So } f''(x) = 0 \text{ has exactly two roots in } \left(\frac{-\pi}{2}, \frac{\pi}{2} \right).$$

$$\text{At } x = \frac{\pi}{4}, f'(x) \text{ has a local minima.}$$

8. Ans (A,B,C)

$$I = \int_0^1 (10x^4 - 3x^2 - 1) dx = [2x^5 - x^3 - x]_0^1 = 0$$

Since, $f(x)$ is even hence must have a root in $(-1, 0)$

PART-3 : MATHEMATICS

SECTION-I (ii)

9. Ans (C)

$$(A) f_2(x) = f(f(x)) = f(x) = x$$

$$f_3(x) = f(f_2(x)) = f(x) = x$$

$$\Rightarrow x^3 - 25x^2 + 175x - 375 = 0$$

$$(x - 5)(x^2 - 20x + 75) = 0$$

$$(x - 5)^2(x - 15) = 0$$

$$\Rightarrow x = 5, 15 \Rightarrow \mathbf{Q, S}$$

(B) Range of $f(f(f(x)))$ is $[4, 17] \Rightarrow \mathbf{Q, R, S}$

Domain of $f(x)$ is $[-1, 1]$

$$(C) \text{ if } x \in [-1, 0), f(x) = 8(x + \pi) + 5x + 4x - x = 16x + 8\pi$$

$$f(x) \in [8\pi - 16, 8\pi]$$

$$\text{If } x \in (0, 1], f(x) = 8x + 5x + 4x - x = 16x$$

$$f(x) \in (0, 16] \Rightarrow \mathbf{P, Q, R, S}$$

(D) $x \in [-1, 0]$

$$x + \frac{1+x^2}{2} = -2x$$

$$x^2 + 6x + 1 = 0 \Rightarrow x = \frac{-6 \pm \sqrt{36-4}}{2} = -3 \pm 2\sqrt{2}$$

$$x = 2\sqrt{2} - 3 \Rightarrow |10a| = \lfloor 20\sqrt{2} - 30 \rfloor = 30 - 20\sqrt{2}$$

$$x \in [0, 1]$$

$$x + \frac{1+x^2}{2} = 2x$$

$$1 + x^2 = 2x \Rightarrow x = 1 \Rightarrow |10a| = 10$$

$$|10a| = 10, \lfloor 20\sqrt{2} - 30 \rfloor$$

$$\lfloor |10a| \rfloor = 1, 10 \Rightarrow \mathbf{P, R}$$

10. Ans (C)

$$(A) f_2(x) = f(f(x)) = f(x) = x$$

$$f_3(x) = f(f_2(x)) = f(x) = x$$

$$\Rightarrow x^3 - 25x^2 + 175x - 375 = 0$$

$$(x - 5)(x^2 - 20x + 75) = 0$$

$$(x - 5)^2(x - 15) = 0$$

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(B) Range of $f(f(f(x)))$ is $[4, 17] \Rightarrow \mathbf{Q, R, S}$

Domain of $f(x)$ is $[-1, 1]$

(C) if $x \in [-1, 0)$,

$$f(x) = 8(x + \pi) + 5x + 4x - x = 16x + 8\pi$$

$$f(x) \in [8\pi - 16, 8\pi]$$

$$\text{If } x \in (0, 1], f(x) = 8x + 5x + 4x - x = 16x$$

$$f(x) \in (0, 16] \Rightarrow \mathbf{P, Q, R, S}$$

(D) $x \in [-1, 0]$

$$x + \frac{1+x^2}{2} = -2x$$

$$x^2 + 6x + 1 = 0 \Rightarrow x = \frac{-6 \pm \sqrt{36-4}}{2} = -3 \pm 2\sqrt{2}$$

$$x = 2\sqrt{2} - 3 \Rightarrow |10a| = \lfloor 20\sqrt{2} - 30 \rfloor = 30 - 20\sqrt{2}$$

$$x \in [0, 1]$$

$$x + \frac{1+x^2}{2} = 2x$$

$$1 + x^2 = 2x \Rightarrow x = 1 \Rightarrow |10a| = 10$$

$$|10a| = 10, \lfloor 20\sqrt{2} - 30 \rfloor$$

$$\lfloor |10a| \rfloor = 1, 10 \Rightarrow \mathbf{P, R}$$

11. Ans (C)

Tangent at point P(x, y), is $y - y = f'(x)(x - x)$

Q; (0, y - x f'(x))

Then, PQ = OQ

$$\Rightarrow x^2 + x^2 (f'(x))^2 = y^2 + x^2 (f'(x))^2 - 2xy (f'(x))$$

$$\Rightarrow x^2 = y^2 - 2xy \cdot f'(x)$$

$$\text{or } dy/dx = \frac{y^2 - x^2}{2xy}, \text{ put } y = tx$$

$$\Rightarrow t + x \frac{dt}{dx} = \frac{dy}{dx}$$

$$\therefore t + x \frac{dt}{dx} = \frac{t^2 - 1}{2t}$$

$$\Rightarrow x \frac{dt}{dx} = \frac{-(1+t^2)}{2t} \Rightarrow \frac{2t}{1+t^2} dt = -\frac{dx}{x}$$

$$\Rightarrow \frac{c}{x} = 1 + \frac{y^2}{x^2} \Rightarrow x^2 + y^2 - cx = 0$$

$$\therefore C_1 : x^2 + y^2 - x = 0$$

$$C_2 : x^2 + y^2 + x = 0$$

C_1 and C_2 touch externally $\Rightarrow$ number of common tangents = 3.

$$C_3 : (x - 5)^2 + y^2 = 5^2$$

Tangent, $bx + ay - ab = 0$, length of perpendicular from centre = radius.

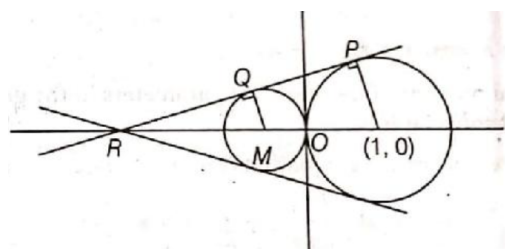
$$\Rightarrow |5b - ab| = 5\sqrt{a^2 + b^2}$$

$$\Rightarrow a^2b^2 - 10ab^2 = 25a^2$$

$$\Rightarrow ab^2 - 10b^2 = 25a$$

$$C_1 : x^2 + y^2 - 2x = 0$$

$$C_2 : x^2 + y^2 - Cx = 0$$



$$\angle QRM = 30^\circ$$

$$\sin 30^\circ = \frac{1}{1 + OR} \Rightarrow OR = 1$$

12. Ans (A)

Tangent at point P(x, y), is $y - y = f'(x)(x - x)$

Q; (0, y - x f'(x))

Then, PQ = OQ

$$\Rightarrow x^2 + x^2 (f'(x))^2 = y^2 + x^2 (f'(x))^2 - 2xy (f'(x))$$

$$\Rightarrow x^2 = y^2 - 2xy \cdot f'(x)$$

$$\text{or } dy/dx = \frac{y^2 - x^2}{2xy}, \text{ put } y = tx$$

$$\Rightarrow t + x \frac{dt}{dx} = \frac{dy}{dx}$$

$$\therefore t + x \frac{dt}{dx} = \frac{t^2 - 1}{2t}$$

$$\Rightarrow x \frac{dt}{dx} = \frac{-(1+t^2)}{2t} \Rightarrow \frac{2t}{1+t^2} dt = -\frac{dx}{x}$$

$$\Rightarrow \frac{c}{x} = 1 + \frac{y^2}{x^2} \Rightarrow x^2 + y^2 - cx = 0$$

$$\therefore C_1 : x^2 + y^2 - x = 0$$

$$C_2 : x^2 + y^2 + x = 0$$

C_1 and C_2 touch externally $\Rightarrow$ number of common tangents = 3.

$$C_3 : (x - 5)^2 + y^2 = 5^2$$

Tangent, $bx + ay - ab = 0$, length of perpendicular from centre = radius.

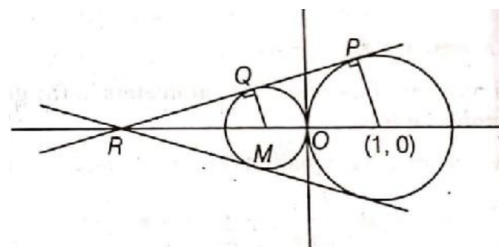
$$\Rightarrow |5b - ab| = 5\sqrt{a^2 + b^2}$$

$$\Rightarrow a^2b^2 - 10ab^2 = 25a^2$$

$$\Rightarrow ab^2 - 10b^2 = 25a$$

$$C_1 : x^2 + y^2 - 2x = 0$$

$$C_2 : x^2 + y^2 - Cx = 0$$



$$\angle QRM = 30^\circ$$

$$\sin 30^\circ = \frac{1}{1 + OR} \Rightarrow OR = 1$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (2021.00)

Given $f(f(x)) = x$, $x \in [0, 1]$

and $f(0) = 1$

$$\text{Let } I = \int_0^1 (x - f(x))^{2020} dx \quad \dots\dots\dots(1)$$

substitute $x = f(y) \Rightarrow dx = f'(y)dy$

also $f(f(x)) = x \Rightarrow f(f(0)) = 0$

$\Rightarrow f(1) = 0$

$$\begin{aligned} I &= \int_1^0 (f(y) - f(f(y)))^{2020} f'(y) dy \\ &= - \int_0^1 (f(y) - y)^{2020} f'(y) dy \\ &= - \int_0^1 (x - f(x))^{2020} f'(x) dx \quad \dots\dots\dots(2) \end{aligned}$$

add (1) & (2)

$$2I = \int_0^1 (x - f(x))^{2020} (1 - f'(x)) dx$$

$x - f(x) = t \Rightarrow (1 - f'(x)) dx = dt$

$$2I = \int_{-1}^1 t^{2020} dt = 2 \int_0^1 t^{2020} dt$$

$$I = \left(\frac{t^{2021}}{2021} \right)_0^1 = \frac{1}{2021}$$

2. Ans (0.50)

$$\lim_{n \rightarrow \infty} n^2 \int_0^{\frac{1}{n}} x^{2019x+1} dx \quad \infty \cdot 0 \text{ form}$$

$$= \lim_{n \rightarrow \infty} \frac{\int_0^{\frac{1}{n}} x^{2019x+1} dx}{\left(\frac{1}{n^2}\right)}, \frac{0}{0} \text{ form}$$

$$= \lim_{n \rightarrow \infty} \frac{\left(\frac{1}{n}\right)^{\frac{2019}{n}+1} \left(-\frac{1}{n^2}\right)}{\left(\frac{-2}{n^3}\right)}, \text{L-hospital rule}$$

$$= \frac{1}{2} \lim_{n \rightarrow \infty} \left(\frac{1}{n}\right)^{\frac{2019}{n}}$$

$$= \frac{1}{2} \lim_{n \rightarrow \infty} e^{\frac{2019}{n} \ln\left(\frac{1}{n}\right)}$$

$$= \frac{1}{2} \lim_{n \rightarrow \infty} e^{\frac{-2019 \ln(n)}{n}}$$

$$= \frac{1}{2} \lim_{n \rightarrow \infty} e^{-\frac{2019}{n}} = \frac{1}{2} e^0 = \frac{1}{2}$$

3. Ans (8.00)

ΔOPQ

$$\frac{OQ}{\sin(\pi - \theta)} = \frac{OP}{\sin\left(\frac{\pi}{2} - \theta\right)}$$

$$OQ = OP \tan \theta \quad \dots (1)$$

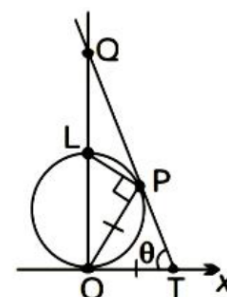
$$\frac{OP}{4} = \sin 2\theta$$

$$\Rightarrow OP = 4 \sin 2\theta \quad \dots\dots(2)$$

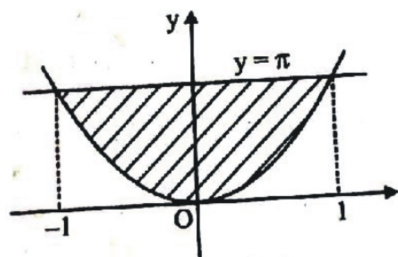
From (1), (2)

$$OQ = 8 \sin^2 \theta ;$$

$$\lim_{\theta \rightarrow \pi/2} OQ = 8$$



4. Ans (4.00)



$$f(x) = \frac{\pi}{2} - \sin^{-1} \left(\frac{2x}{1+x^2} \right) = \frac{\pi}{2} - 2\tan^{-1} x$$

$$g(x) = \frac{\pi}{2} - \tan^{-1} \left(\frac{2x}{x^2-1} \right) = \frac{\pi}{2} + 2\tan^{-1} x$$

$$y = f(x) + g(x) = \pi$$

$$y = \pi x^2$$

$$\text{Required area} = 2\pi - 2 \int_0^1 \pi x^2 dx$$

$$A = 2\pi - 2\pi \cdot \frac{1}{3} = \frac{4\pi}{3} \approx 4.19$$

5. Ans (3.00)

$$c \times a = b \Rightarrow |c \times a| = |b|$$

$$\Rightarrow |c| |a| \sin \theta = 3,$$

$$|c| = \frac{3}{2 \sin \theta} |c - a|^2 = |c|^2 + |a|^2 - 2c \cdot a$$

$$= |c|^2 + 4 - 2|c| \cdot |a| \cos \theta$$

$$= \frac{9}{4 \sin^2 \theta} + 4 - 2 \cdot \frac{3}{2 \sin \theta} \cdot 2 \cdot \cos \theta$$

$$= 4 + \frac{9}{4} \cos^2 \theta - 6 \cot \theta$$

$$= \frac{9}{4} + \left(\frac{3}{2} \cot \theta - 2 \right)^2$$

$$|c - a|^2 \geq \frac{9}{4}$$

$$\Rightarrow |c - a| \geq \frac{3}{2}$$

$$\Rightarrow 2|c - a| \geq 3$$

$$\therefore \text{Min. of } 2|c - a| = 3$$

6. Ans (2.00)

$$x^2 + y^2 + z^2 = \text{Square of distance from origin}$$

$$4\sin^2 t + 4\cos^2 t + 9t^2 = 4 + 9t^2$$

Which is shortest at $t = 0$

$$\Rightarrow \text{Shortest distance} = 2$$